

WORK EXPERIENCE AND RESEARCH

Research Assistant

Apr. 2024 - Present

Nepal Applied Mathematics and Informatics Institute for research
Supervisor: Dr. Binod Bhattarai

Lalitpur, Nepal

- Working on research projects focused on **trustworthy machine learning**.
- Developed an **OOD detection method** for pathology segmentation in volumetric medical images, specifically MRI and CT scans, based on **self-supervised learning** with synthetic anomalies, leading to a **5th-rank** in the MICCAI 2024 challenge.

Undergraduate Research Intern

Aug. 2023 - Mar. 2024

Nepal Applied Mathematics and Informatics Institute for research
Supervisor: Dr. Binod Bhattarai

Lalitpur, Nepal

- Worked on an **Active Learning framework** to determine the most effective **uncertainty-based sampling technique** to address dataset imbalance in a multi-class object detection task.
- Trained deep learning models and conducted an ablation study for detecting gait abnormalities in warehouse environments.

EDUCATION

Pulchowk Campus - Institute of Engineering, Tribhuvan University

Nov. 2019- Apr. 2024

B.E in Electronics, Communication and Information

- Achieved Rank 104 in the university entrance examination out of nearly 18,000 candidates - **top 0.6%**, and received the scholarship provided by the Government of Nepal covering full tuition expenses
- Graduated with **75.73%**, ranked in the top 5% of students, and as the **top-performing female student** in the program.

PUBLICATIONS

- Chhetri, A., Shrestha, P., Rana, R., Gyawali, P., & Bhattarai, B. (2026). **When Confidence Lacks Concepts: Interpretable OOD Detection via Representation Perturbations**. (*Preprint*)
- Chhetri, A., Korhonen, J., Gyawali, P., & Bhattarai, B. (2025). **NERO: Explainable Out-of-Distribution Detection with Neuron-Level Relevance in Gastrointestinal Imaging**. *In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2025*

SELECTED ACADEMIC PROJECTS

Improving Adversarial Image Robustness For Preventing Image Manipulation

Oct. 2023 - Apr. 2024

Capstone Project

[Link to Repo](#)

- Developed a **JPEG-resistant image immunization pipeline** to prevent text-guided AI manipulation, specifically for text-guided diffusion models.
- Investigated the impact of text prompts on the **success of adversarial attacks** against **text-guided diffusion models**.

Real-Time Visual Element Generation Using GANs

Dec. 2022 - Apr. 2023

Academic Project

[Paper](#) | [Link to Repo](#)

- Developed a **real-time neural rendering system** for generating realistic grass in video games.
- Experimented with existing architectures, performed result analysis, and designed new architectures that achieved both visual fidelity and low inference time.

TEACHING EXPERIENCE

Fifth Annual Nepal AI School - NAAMII

27th Dec., 2024 - 6th Jan., 2025

Teaching Assistant

Certificate

- Performed live demonstrations of adversarial attacks in image models and LLMs to an audience of 100 students under the supervision Prof. Yash Raj Shrestha (University of Lausanne, Switzerland).
- Assisted in lab sessions on Geometric Deep Learning, Active Learning, and Self-Supervised Learning.

AWARDS AND HONORS

Nepal-US Hackathon - Category Winner (Innovative) Nepali Leaders Network	March 2026
Eastern European Machine Learning - Full Scholarship ANNT, Google DeepMind, University of Sarajevo	July 2025
Mentee, Women Leaders Mentorship Programme, <i>EU in Nepal</i> Mentor: H.E. Veronique Lorenzo , EU Ambassador to Nepal, 4% acceptance rate	Dec. 2024–October 2025
Medical OOD Analysis Challenge, 5th Place , Pixel-level Task MICCAI 2024	October 2024
Hacking for Humanity 2023, 2nd Runner Up Shequal Foundation, Nepal's largest all female hackathon	February 2023
GritFeat AI Health Hackathon, 1st Position GritFeat Solutions	January 2023
ABU-Robocon 2022, 2nd Runner Up Asia-Pacific Robot Contest	August 2022
3rd Annual Nepal AI School - Full Scholarship Nepal Applied Mathematics and Informatics Institute for Research	December 2021
Mahatma Gandhi Scholarship Embassy of India	2017-2018

SERVICE

Reviewer

- 29th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2026)

Volunteering

03/2026 – ongoing	Co-Operations Lead (AI Safety Nepal) : Talks and workshops on AI and AI safety for general public and schools.
5/2025 – 09/2025	Web and Publicity Chair (DEMI workshop at MICCAI 2025) : managing the online presence and public communication.
10/2024 – 12/2024	Organizing and Admission Committee (Fifth Annual Nepal AI School) : Managed logistics, including stakeholder coordination, task division, and application evaluation
08/2024	AI Trainer, Right to Information (Sancharika Samuha, The Asia Foundation, USAID Nepal) : Supported mid-career journalists in eastern Nepal on enhancing their understanding of AI technology.
11/2023	Children in Technology, 2023 (LOCUS 2023) : Awareness program on internet safety and cyber law to underserved rural communities in eastern Nepal.

Student Activities

- Event Manager, IT Club - Pulchowk Campus, 2021-2023
- Student Member, Robotics Club - Pulchowk Campus, 2019-2022